

# TASK-4 REPORT

## Cognifyz Machine Learning Internship

**Task:** Location-Based Analysis

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### 1. Objective

The aim of this task is to analyze restaurants based on their **location** using the given dataset. The analysis focuses on how cities and localities affect restaurant distribution, cuisines, ratings, and price ranges.

### 2. Dataset

- **Dataset Name:** Restaurant.csv
- **Source:** Cognifyz Technologies

The dataset contains details of **9,551 restaurants**, including city, locality, coordinates, cuisines, cost, and ratings.

### 3. Method Used

- The dataset was loaded and cleaned
- Missing latitude and longitude values were removed
- Restaurant locations were visualized using **Folium** with **Marker Clustering**
- Restaurants were grouped by city and locality
- Average ratings and price ranges were calculated for each area
- Cuisine types were analyzed to find the most popular ones

### 4. Key Findings

- Most restaurants are concentrated in **New Delhi**, followed by Gurgaon and Noida
- **Inner City** has the highest average rating and highest price range
- Popular cuisines include **North Indian, Chinese, and Fast Food**
- Restaurant distribution is much higher in metropolitan areas than in smaller cities

### 5. Results

The analysis clearly shows that restaurant density, pricing, and cuisine popularity vary strongly by location.

All charts and the interactive map were generated successfully.

### 6. Conclusion

The location-based analysis was completed successfully.

It demonstrates how geographical factors influence restaurant availability, pricing, and customer ratings.

### 7. Future Scope

- Create heatmaps for better visualization
- Compare restaurant trends across countries
- Build an interactive dashboard
- Use clustering techniques to group similar cities